

# The two Banach-valued ergodic questions of arXiv:0712.1740

Later affirmative answers

Autonomous mathematical discovery run `fa_banach_001`

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**Status.** Literature already answered: both questions have affirmative answers, with the standard hypotheses stated below.

## 1 Source questions

In the unnumbered “Questions” subsection of the outlook to [1, PDF pp. 19–20], Gruber, Lenz, and Veselić ask:

1. whether their Banach-space-valued ergodic theorem over  $\mathbb{Z}^d$  extends, by a similar argument, to finitely generated discrete groups of polynomial growth; and
2. whether there is one Banach-space-valued ergodic theorem applicable to both  $\mathbb{Z}^d$ - and  $\mathbb{R}^d$ -equivariant models, thereby avoiding a separate treatment of discrete and continuous group actions.

## 2 Identification of the later answers

### Question 1: polynomial-growth groups

Lenz, Schwarzenberger, and Veselić give the requested extension in [2]. Their Assumption 2.5 works on a finitely generated amenable group with a suitable tiling Følner sequence, pattern frequencies, a Banach target space, and a colouring-invariant almost-additive set function. Their Remark 2.6 explicitly states that every group of polynomial volume growth fits this framework. Theorem 3.1 then proves convergence in the Banach-space norm and gives an explicit error estimate in terms of the boundary term and pattern frequencies.

Thus the first question has a full affirmative answer. The theorem covers the requested polynomial-growth class and preserves the two characteristic features of the source method: almost-additive Banach-valued functions and quantitative control by tilings and frequencies.

### Question 2: one discrete/continuous theorem

Pogorzelski subsequently treats a second-countable, locally compact, unimodular amenable group  $G$  with its Haar measure [3]. The paper notes that every discrete group and every abelian group is unimodular. Its Theorem 5.7, “Mean ergodic theorem for set functions,” proves Banach-norm convergence of bounded almost-additive functions along strong Følner sequences, under a weak compactness condition on the translation orbits and the stated compatibility with the group action.

This is precisely one abstract framework containing both cases in the second question:

$$G = \mathbb{Z}^d \quad \text{with counting Haar measure,} \quad G = \mathbb{R}^d \quad \text{with Lebesgue Haar measure.}$$

Theorem 5.7 therefore gives an affirmative answer at the level asked by the source: the discrete/continuous distinction is absorbed into the choice of locally compact group and Haar measure. The later paper also supplies pointwise and integrated-density-of-states applications, but those are not needed for the identification.

### 3 Scope and confidence

The answer to Question 2 is an abstract unification, not a claim that every individual model in the two earlier papers automatically satisfies every hypothesis of Theorem 5.7. Applications must still verify almost additivity, measurability, the boundary condition, and weak compactness. This is the ordinary scope of the requested theorem rather than a gap in the answer.

The identification is high confidence. The first later paper explicitly singles out polynomial volume growth immediately before its Banach-valued Theorem 3.1. The second paper explicitly advertises Theorem 5.7 as a theorem for locally compact unimodular amenable groups, and its preliminaries explicitly include both discrete groups and abelian groups. We use the corrected arXiv version 2 of [3].

### References

- [1] M. J. Gruber, D. H. Lenz, and I. Veselić, *Uniform existence of the integrated density of states for combinatorial and metric graphs over  $\mathbb{Z}^d$* , arXiv:0712.1740v1 (2007).
- [2] D. Lenz, F. Schwarzenberger, and I. Veselić, *A Banach space-valued ergodic theorem and the uniform approximation of the integrated density of states*, Geom. Dedicata 150 (2011), 1–34; arXiv:1003.3620v2.
- [3] F. Pogorzelski, *Almost-additive ergodic theorems for amenable groups*, J. Funct. Anal. 265 (2013), no. 8, 1615–1666, with corrigendum/addendum; arXiv:1211.2089v2.